

Unsupervised Image-Based Prompting for Lesion Matching Across Mammography Views

Santiago Pires¹, Amir Hadjifaradji¹, Joep Stevens¹, Ana San Roman Gaitero¹, Jaap Kroes¹, Nico Karssemeijer¹, and Alejandro Rodriguez Ruiz¹

¹ScreenPoint Medical, Nijmegen, Netherlands



Purpose

Radiologists will use information from the **craniocaudal (CC)** and mediolateral **oblique (MLO)** views to **correlate and assess suspicious findings** [1]. This challenging task has been reported to have an error rate of 23% [2].

Existing CAD systems and most AI models still analyse each view independently. Solving such problems could help boost performance and **assist radiologists**.

Our contributions are the following:

- Extending the **DINOv2** [3] framework for processing low-resolution CC and MLO images simultaneously and **reinforcing a masked-patch prediction objective** that works as a **proxy task for matching**.
- A **promptable method** for using the model's **patch representations** to retrieve the most similar region in the corresponding view.

Material & Methods

Proxy task: We mask patches in one view (e.g., CC) and let the model infer their representations from the opposite view (e.g., MLO), encouraging use of complementary information (applied CC→MLO and MLO→CC).

Data: Approximately **100,000 paired CC-MLO screening mammograms**.

Inputs: Each CC-MLO pair is **downsampled and concatenated** before being fed to the **DINOv2 ViT**.

Loss: The **iBOT** objective is **adjusted** to encourage **cross-view embedding alignment**.

Results

Task: A **source lesion** is manually delineated (prompted) in one view; the goal is to locate its counterpart in the other view.

Linking: Collect all **patch tokens** inside the source lesion and compute **cosine similarity** against every token in the cross-view image.

Cohort: **Malignant and benign** cases where **both views contain a matching lesion**.

Metric: **Correct cross-view link rate** = % of predictions whose **center falls inside the ground-truth (GTR) lesion footprint**.

Performance: On a held-out set of **938 paired lesions**, we achieve **78.0%** correct links (**95% CI: 75.3–80.6**).

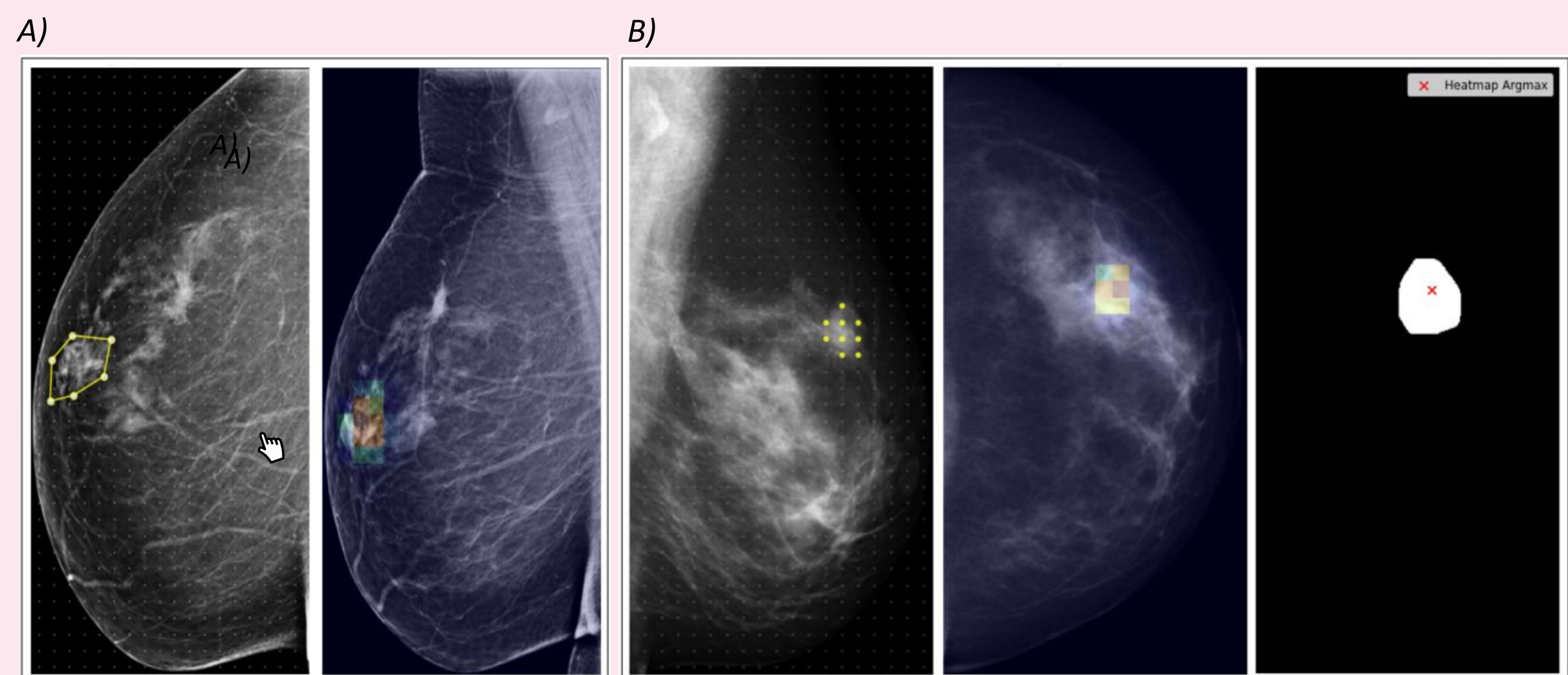


Figure 1. a) Inference: CC view with a manually prompted lesion contour and the resulting heatmap over the MLO view. b) Evaluation: using existing GTRs in the MLO view with CC heatmap and peak prediction in GTR.

Conclusion

Result: **78.0%** correct cross-view links on **938** lesions, demonstrating effective correspondence learning.

Label-free: **No lesion labels required**; learning signal comes from the **cross-view masked-patch proxy**.

Promptable: **Any user-selected region** can be linked across views at inference.

Outlook: The learned representation **may transfer** to improve **cancer classification** and related tasks.

References

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